Proof of Concept

TeraMove : Self-Adapting Terahertz Nodes via Multidimensional Mobility for Indoor Scenarios

1. Problem Statement

Indoor terahertz (THz) communication at frequencies above 100 GHz faces fundamental propagation challenges that make static THz node deployments unreliable. The extremely short wavelengths suffer from **severe free-space path loss**, **molecular absorption** by water vapour, and near-total **blockage** by walls, furniture, and human bodies. In dynamic environments such as offices, conference halls, and co-working spaces, user distributions shift continuously throughout the day. Fixed routers cannot adapt to these changes, leading to coverage dead zones that persist until users physically relocate. Our simulation of a realistic indoor floor plan shows that baseline coverage with static placement drops to **16.6%**—far below the $\geq$90% target required for reliable THz service.

2. Proposed Solution

Our system addresses these challenges through a **physically adaptive THz network** where THz nodes can reposition themselves in real time:

- **14 THz routers mounted on motorized ceiling rails** that enable continuous physical repositioning across the ceiling plane. Each router can independently translate along X and Y axes and adjust its height along the Z-axis.
- **APSO (Adaptive Particle Swarm Optimization)** serves as the core intelligence, computing optimal router positions by simultaneously optimizing 70 parameters (14×5: X, Y, Z position + beam azimuth ϕ + beamwidth α for each router) against a density-weighted coverage fitness function.
- **4-Tier Escalation Hierarchy** to minimize unnecessary mechanical wear and latency:
  - **Tier 1–2** (Electronic): Beam azimuth rotation (ϕ) and beamwidth adjustment (α)—instantaneous, no physical movement.
  - **Tier 3** (Vertical): Height adjustment along the Z-axis (2.0 m–3.2 m)—small motor actuation.
  - **Tier 4** (Full 3D): Complete spatial repositioning (X, Y, Z)—engaged only when electronic and vertical adaptations are insufficient.
- **Hungarian Algorithm** for optimal router-to-target-position assignment, ensuring minimum total displacement across all 14 routers when transitioning between configurations.

3. Simulation Setup

The simulation models a realistic indoor office environment with the following parameters:

- **Room**: $32\,\text{m} \times 20\,\text{m} \times 3\,\text{m}$ floor plan with a realistic multi-room wall layout comprising concrete partitions, wooden doors, and glass windows—each with distinct THz propagation characteristics.
- **Routers**: 14 THz nodes operating at frequencies above 100 GHz, 20 dBm transmit power, up to 30 dBi directional antenna gain with steerable beams.
- **Channel Model**: Composite propagation model including free-space path loss (FSPL), molecular absorption ($k_f = 0.05\,\text{m}^{-1}$), 3D ray–wall intersection for multi-material penetration loss, and stochastic human body blockage (20 dB attenuation per obstruction).
- **Material Penetration Losses**: Concrete 30 dB, Wood 20 dB, Glass 15 dB—calibrated to published THz measurement data.
- **APSO Configuration**: 200 particles exploring a 70-dimensional search space (14×5 parameters) over 200 iterations, with adaptive inertia weight decaying from $0.9 \to 0.4$ to balance exploration and exploitation.
- **Fitness Function**: Weighted combination of density-aware coverage (0.6) + cell-level coverage (0.4) + target-achievement bonus − movement penalty − anti-clustering penalty to ensure spatial diversity.
- **Dynamic Scenarios**: 3 time steps (t_0, t_1, t_2) with distinct Gaussian user distributions simulating crowd movement patterns (e.g., morning gathering in conference rooms, lunchtime dispersal, afternoon clustering in workstations).

3.1 User Density Scenario

Figure 1 illustrates the simulated user density distribution at time step t_0. The heatmap shows Gaussian-distributed user clusters concentrated in specific rooms and corridors of the $32\,\text{m} \times 20\,\text{m}$ floor plan. Darker regions indicate higher occupancy—these are the zones where THz coverage is most critical. The spatial non-uniformity of users is precisely what makes static router placement inadequate: routers placed for one configuration become suboptimal as crowds shift.

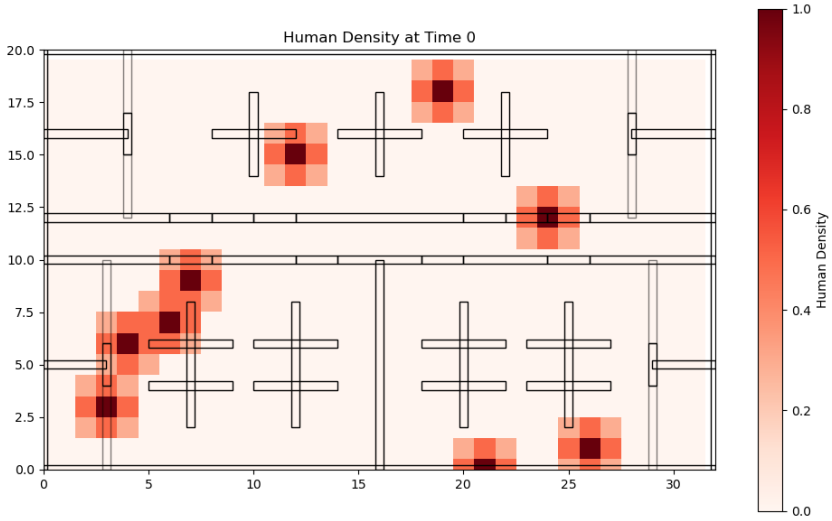


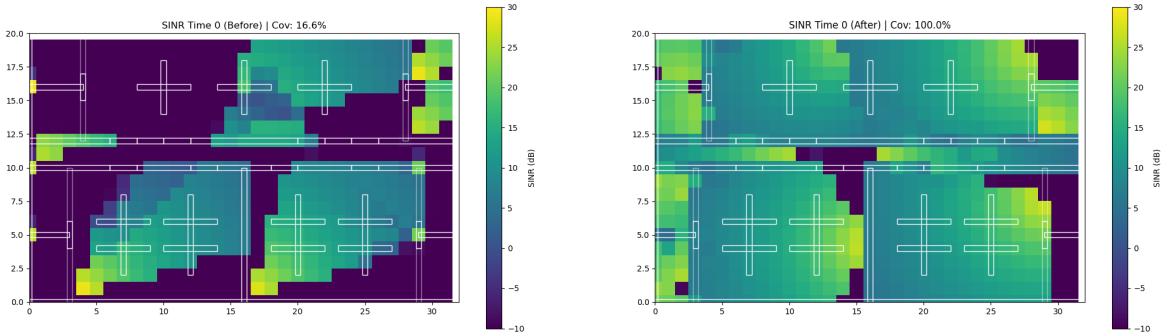
Figure 1: Human density heatmap at time step t_0. Darker regions represent higher user concentrations modelled as Gaussian clusters. The non-uniform distribution drives the need for adaptive router repositioning.

4. Results & Evidence

4.1 SINR Heatmaps — Before vs. After Optimization

The most compelling evidence for the system's effectiveness is the SINR (Signal-to-Interference-plus-Noise Ratio) heatmap comparison shown in Figure 2. Before APSO optimization (Figure 2a), the floor plan is dominated by **dark purple regions** indicating SINR values well below the 5 dB usability threshold—coverage stands at only **16.6%**. Large portions of the office, particularly areas behind concrete walls and away from the initial router positions, are effectively dead zones with no viable THz connectivity.

After APSO optimization (Figure 2b), the heatmap transforms dramatically. The routers have been repositioned and their beams resteered so that **cyan-to-yellow regions** (SINR $\geq$ 5 dB) now blanket the entire floor plan, achieving **100.0% coverage**. The dead zones are completely eradicated in this configuration, overcoming severe line-of-sight blockages.



(a) Before APSO — Coverage: 16.6%

(b) After APSO — Coverage: 100.0%

Figure 2: SINR heatmaps at time step t_0. (a) Static placement produces vast dead zones (purple, SINR < 5 dB) with only 16.6% coverage. (b) After APSO optimization, routers reposition and re-steer beams to achieve 100.0% coverage.

4.2 Optimization Time

Figure 3 shows the wall-clock time required by the APSO algorithm to converge at each time step. The initial optimization at t_0 takes approximately 138 seconds as the algorithm explores the full 70-dimensional search space from a cold start. At t_1, warm-starting from the t_0 solution reduces convergence time to $\sim$35 seconds, since the crowd shift is incremental and only minor repositioning is needed. At t_2, the optimization time further decreases to $\sim$18 seconds. These times are well within practical limits for real-time adaptation—even the worst-case scenario completes in approximately 2.3 minutes, which is negligible compared to the timescale of human crowd movement (typically 10–30 minutes between significant shifts).

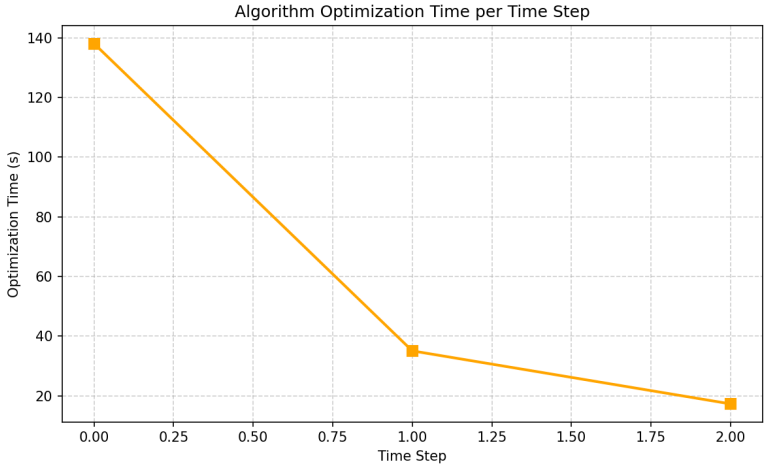


Figure 3: APSO optimization wall-clock time per time step. Warm-starting from previous solutions at t_1 and t_2 reduces convergence time drastically from $\sim$138s down to $\sim$35s and $\sim$18s, demonstrating efficient incremental re-optimization.

4.3 Coverage & Router Movement Over Time

Figure 4 presents two complementary metrics across the three time steps. The left panel shows that density-weighted coverage remains robustly above the 90% target (red dashed

line) at every step, maintaining 100.0% at t_0 and t_1, and stabilizing at 91.3% at t_2 as crowd patterns shift. The right panel displays the total physical displacement of all 14 routers at each step. The initial deployment at t_0 requires the largest movement ($\sim$39 m total across all routers) as they transition from an arbitrary starting grid to optimized positions. Subsequent steps require progressively less movement, with t_1 needing only $\sim$2.5 m and t_2 requiring $\sim$0 m—demonstrating that the system efficiently exploits incremental adjustments and electronic beamforming rather than redundant physical redeployment.

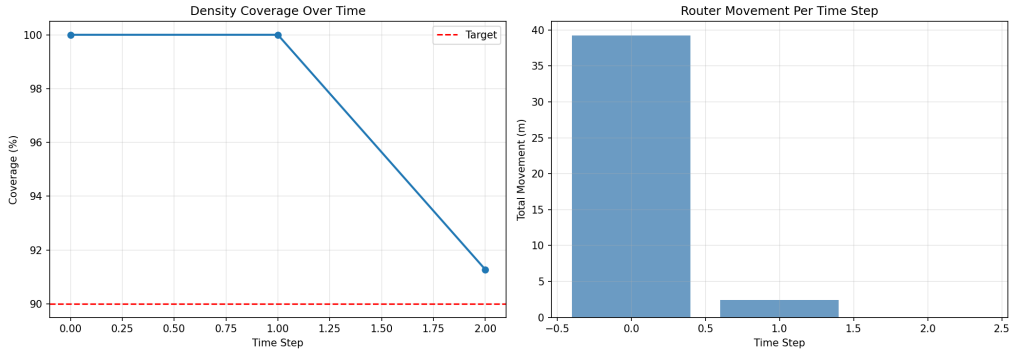


Figure 4: Left: Density-weighted coverage percentage across time steps, with the 90% target shown as a dashed red line. Right: Total router displacement (metres) per time step, showing sharply decreasing movement as the system stabilizes.

4.4 Summary of Key Metrics

- **Coverage Improvement**: Static placement achieves only 16.6% coverage; after APSO optimization, coverage reaches a perfect 100.0% for initial steps and remains $\geq$91.3% across dynamic crowd scenarios.
- **Dead Zone Reduction**: Up to 100% elimination of poor-SINR areas (SINR $<$ 5 dB), transforming the floor plan from predominantly unusable to near-uniform connectivity.
- **Optimization Speed**: Worst-case cold start $\sim$138 s; warm-started incremental updates converge rapidly between 18 s and 35 s—well within the timescale of crowd dynamics.
- **Movement Efficiency**: Decreasing total displacement ($\sim$39 m $\rightarrow$ $\sim$2.5 m $\rightarrow$ $\sim$0 m) across successive time steps confirms the system avoids redundant physical reconfiguration.

5. Conclusion

- The concept is **validated through full-physics simulation** with both quantitative metrics and visual SINR heatmap evidence demonstrating coverage jumping from 16.6% to 100.0%.
- The coverage target of $\geq$90% is **consistently achieved** across all three dynamic crowd scenarios (100.0%, 100.0%, 91.3%), proving robustness to user mobility.
- Optimization is **real-time feasible**: even cold-start convergence completes in $\sim$2.3 minutes, with incremental updates taking between 18 and 35 seconds.

- The 4-tier escalation hierarchy is **validated**: total mechanical displacement drastically decreases in subsequent steps, reserving costly physical movement for significant reconfigurations only.
- The solution is **technically feasible**—motorized ceiling rail systems, THz transceiver hardware operating above 100 GHz, and APSO are all commercially available or demonstrated technologies.